

Project Development Phase

Preprocessing Steps & Business Questions

Field	Details
Team Lead	Govindu Kavya Sri
Team Members	Eswari Gayatri Budumuru, Kiladi Kalavathi, Adaka Mounika, Panditi Jyostna
Domain	Retail / Cosmetics Analytics
Tools Used	Python, Microsoft Excel, Tableau Public
Dataset	Cosmetics product dataset — 1,472 records, 11 attributes
Published Dashboard	public.tableau.com/app/profile/govindu.kavya.sri/viz/cosmeticproject_17872016666220/Dashboard1

Project Context

The cosmetics industry is one of the fastest-growing and highly competitive sectors, driven by changing consumer preferences, diverse skin types, brand innovation, and evolving pricing strategies. With increasing demand for personalized beauty products, companies rely on data-driven insights to understand market trends, consumer behavior, and product performance.

This project, “Cosmetic Insights: Navigating Cosmetics Trends and Consumer Behavior with Tableau,” focuses on analyzing a cosmetics dataset to uncover meaningful patterns and actionable insights. The dataset includes key attributes such as brand, product category, price, rank, and skin type suitability (Dry, Oily, Combination, Normal, and Sensitive).

This project is designed to help businesses and analysts understand cosmetics market trends and consumer preferences. By analyzing brand performance, pricing patterns, product distribution, and skin-type coverage, the dashboard enables stakeholders to identify high-performing products, target specific customer segments, and optimize marketing and product strategies.

Scenario 1: Monitoring Consumer Preferences

In a real-time scenario, imagine receiving an alert indicating a concerning trend in consumer preferences, such as a significant decline in interest in certain cosmetic products or ingredients. Using the Cosmetic Insights data, we can promptly assess the extent and potential impact of this trend, identify contributing factors, and deploy immediate interventions to adapt product offerings and marketing strategies. Whether it's through targeted promotional campaigns, adjustments in product formulations, or personalized

recommendations, real-time analysis enables agile decision-making and proactive measures to meet evolving consumer needs.

Scenario 2: Addressing Product Concerns

In the event of identifying widespread product concerns, such as negative reviews or safety issues associated with specific cosmetic items, real-time access to Cosmetic Insights data enables swift response and management. Cosmetic companies and regulatory bodies can utilize the dataset to gather crucial information about the concerns, including their prevalence, potential impacts on consumer trust, and affected product demographics. By leveraging real-time analytics, they can implement quality control measures, recall products if necessary, and communicate transparently with consumers to address their concerns and maintain brand integrity.

Scenario 3: Predictive Analysis and Product Innovation

Leveraging predictive analytics capabilities, Cosmetic Insights empowers companies to anticipate and respond to emerging trends and consumer preferences in the beauty industry. By analyzing historical data and identifying predictive indicators, companies can proactively innovate new products, adjust existing formulations, and tailor marketing strategies to meet evolving consumer demands. Real-time monitoring of market trends, consumer feedback, and competitor activities enables timely interventions, product innovation, and strategic decision-making to stay ahead in a competitive market landscape.

Data Preprocessing Steps

Before any chart was built, the raw cosmetics product extract (1,472 rows) was cleaned and enriched using Python 3 with the Pandas library.

Data Quality Checks Applied

- Row-count and column-type check: confirmed all 11 fields loaded with correct text/number types
- Null and duplicate check: confirmed no fully duplicate rows and identified 143 ingredient-placeholder rows and 19 zero-rank rows for correction
- Casing consistency check: brand names verified as all-uppercase in the raw file, standardized to title case
- Range check: Rank values checked against the expected 1–5 scale to flag the 0-value rows as unrated rather than real scores

Step	Action	Why It Was Needed
1	Load Raw Extract	Single flat CSV file; every subsequent check has to be built manually
2	Standardize Brand Casing	Prevents the same brand appearing as multiple inconsistent entries in filters and charts
3	Replace Ingredient Placeholders	“No Info” and “Visit the [Brand] boutique” are scrape artifacts, not real ingredient data, and would mislead any future ingredient-level analysis

Step	Action	Why It Was Needed
4	Convert Rank 0 to Missing	A Rank of 0 is not a real rating on the 1–5 scale; leaving it as 0 would distort every average-rating chart
5	Rename Skin-Type Columns	Clear, self-explanatory field names (Combination_Skin, etc.) instead of ambiguous originals
6	Pivot Skin-Type Columns	Five separate binary columns cannot be filtered or charted flexibly; pivoting into Skin_Type / Suitable enables the heatmap and dynamic filtering
7	Export Cleaned File	cosmetics_cleaned.xlsx is the single file Tableau connects to

Business Questions & Visualizations

Q#	Business Question (Persona)	Answered By	Key Insight
Q1	How does average price vary across product categories? (Meera, Retail Manager)	Avg Price by Category chart	Treatment and Eye Cream carry the highest average prices; Cleanser is the most affordable category
Q2	Does a higher price reliably mean a better rating? (Priya, Consumer)	Price vs Rating scatter chart	Price and rating are not strongly correlated — several well-rated products sit in the lower price range
Q3	Which brands are genuinely top-rated within a category? (Arjun, Brand Analyst)	Top Brands by Category chart	Several boutique/niche brands outrank larger mass-market names on average rating
Q4	Which categories best serve which skin types? (Priya, Consumer)	Skin Type Coverage heatmap	Skin-type coverage varies meaningfully by category, revealing gaps for certain skin types in certain categories
Q5	How can a user narrow the whole dashboard to brands they already recognize? (all personas)	Searchable Brand filter + cross-chart filter actions	Selecting a category or brand updates all four charts simultaneously, letting any persona reach a relevant view in a few clicks